



Silver management trainer

PRAXIS makes PERFECT



CONTENTS

<i>0. Introduction</i>	4
<i>1. Companies</i>	4
<i>2. Environment</i>	5
<i>3. Products</i>	6
<i>4. Markets</i>	7
<i>5. Demographic profiles of purchasers</i>	8
<i>6. Commercial distribution</i>	9
<i>7. Wholesalers</i>	10
<i>8. Sales management and physical distribution</i>	10
<i>9. Physical distribution - logistics</i>	13
<i>10. Factory to platforms transportation</i>	14
<i>11. Platforms managed by sales managers</i>	16
<i>12. Platforms to retailers transportation</i>	16
<i>13. Sales force</i>	18
<i>14. Prices</i>	19
<i>15. Advertising</i>	19
<i>16. Scope (rating)</i>	20
<i>17. Brand positioning</i>	21
<i>18. Sales promotion</i>	22
<i>19. Point of purchase advertising</i>	23
<i>20. Retailers</i>	24
<i>21. Shelf space</i>	24
<i>22. Production facilities</i>	25
<i>23. Production capacity</i>	26
<i>24. Maintenance</i>	26

25. Preventive maintenance	27
26. Quality control	28
27. Main production module	28
28. Purchasing and supplies	30
29. Office and management staff	31
30. Direct production personnel	32
31. Work environment	33
32. Selection and training	34
33. Standard cost	35
34. Inventories	35
35. Depreciation expense	35
36. Research and development	36
37. Financing	36
38. Customer payments	37
39. Factoring	37
40. Periods - fiscal year	38
41. Operating cash needed	39
42. Loan and cash & equivalents	40
43. Credit lines	41
44. Financial investments & fixed term deposits	42
45. Investing in other companies (acquisitions)	42
46. Currency exchange	42
47. Income statement and balance sheet	43
48. Demand	43
49. Macroeconomic, financial and production factors	44
50. Market research	45
51. Appendix	47

0. INTRODUCTION

You have been chosen to manage a skincare products company headquartered in the European Union. The company produces and markets sunscreens and/or moisturizing lotions. Your company has to draft a management plan for a one-year period. This plan takes the explicit shape of a set of decisions. These decisions are processed by the Praxis MMT 39 simulator, which calculates the results achieved by each company and then shows results to the participants. Each company is unaware of the decisions and results of the others, except for the information that can be obtained from market research.

Based on these results, participants must draft a new management plan for another year that will give rise to new decisions, which will follow the same procedure as described in the previous paragraph.

The simulation Director will stipulate the duration of the simulation, i.e. for how many “years” a new annual management plan must be prepared. The final results obtained by each company are compared with those from its competitors, the goal being to maximize profits. The role of each company is to apply the knowledge and criteria required in real-world management decision-making.

1. COMPANIES

Initially, the companies start from the same situation, and therefore have identical Balance Sheets. Fixed assets consist in land, buildings and production facilities.

Initial Balance Sheet in euros:

Assets		Liabilities and Owners' Equity	
Fixed Assets	12,000,000	Equity	66,000,000
Inventory	--		
Cash and Equivalents	54,000,000		



All numeric data appearing in this document use the convention of separating thousands by a comma and decimals by a dot. In decisions where decimals can be applied, the simulator accepts up to two decimals.



2. ENVIRONMENT

There are three markets, corresponding to three territories, referred to as A, B and E. Markets A and B are two areas within the European Union and therefore use a common currency, the euro. Market E is an area in a foreign country that can be associated with a U.S. region, whose currency is the dollar. Consequently, before the decisions for each business year are made, the simulation Director will provide the exchange rate between the dollar and the euro at that particular moment for those companies that decide to export.

All the competing companies are geographically located somewhere between A and B and are subject to their country's regulations. Therefore, their Income Statement and Balance Sheet are presented in euros. Market E is an export market.

The demographic and geographical data of the three territories are as follows:

	Population	Surface Area
A	22,000,000	100,000 Km ²
B	18,000,000	95,000 Km ²
E	30,000,000	210,000 Km ²



Each market has a different per capita income, subject to change every year.

Companies in markets A and B have no competition other than themselves. On the other hand, in market E, besides the possible performance of other participant companies, there are also domestic companies competing with those companies that decide to export. These companies are assumed to act as one in their decision-making strategy.



In market E, all decisions involving money are to be expressed in euros, as the simulator applies the corresponding euro-dollar exchange rate.

3. PRODUCTS

Each company can manufacture two products that we will call S and H. Both products are in the personal care sector and have high turnover rates. Participants can identify them as sunscreen (S) and moisturizing lotion (H).

The model behaves just as real markets would for these two particular products, although the intention is not for it to be absolutely exact.

The approximate per capita consumption for each product at the beginning of the simulation is as follows:

	Market A	Market B	Market E
Product S	0.55 units	0.50 units	1.17 units
Product H	1.33 units	1.5 units	2.50 units



Obviously, demand will change according to the marketing strategy employed by each company.

Regarding brand positioning, if a company sells the same product in two or three markets, it can either use the same positioning strategy for all of them, or use a different one for each market, just like in the real world.

Both products have a long shelf life and what is left as inventory at the end of a period is what is first sold in the next one. In any case, inventories will not deteriorate and are not subject to realization.



4. MARKETS

Consumer attitudes and behaviors are directly related to their wealth, as defined by each market's per capita income.

Products S and H are in a sustained maturity phase in markets-territories A, B and E.

In markets A and B there are three distribution channels that we will call T, S and G. In market E there is only one type of channel, G. The number of points of sale are specified in the following table:

		A	B	E
T	Traditional retailers	2,160	1,080	
S	Supermarkets	1,620	1,080	
G	Hypermarkets	35	30	200



Traditional retailers are mainly pharmacies and perfume stores, and the associated sections in department stores.

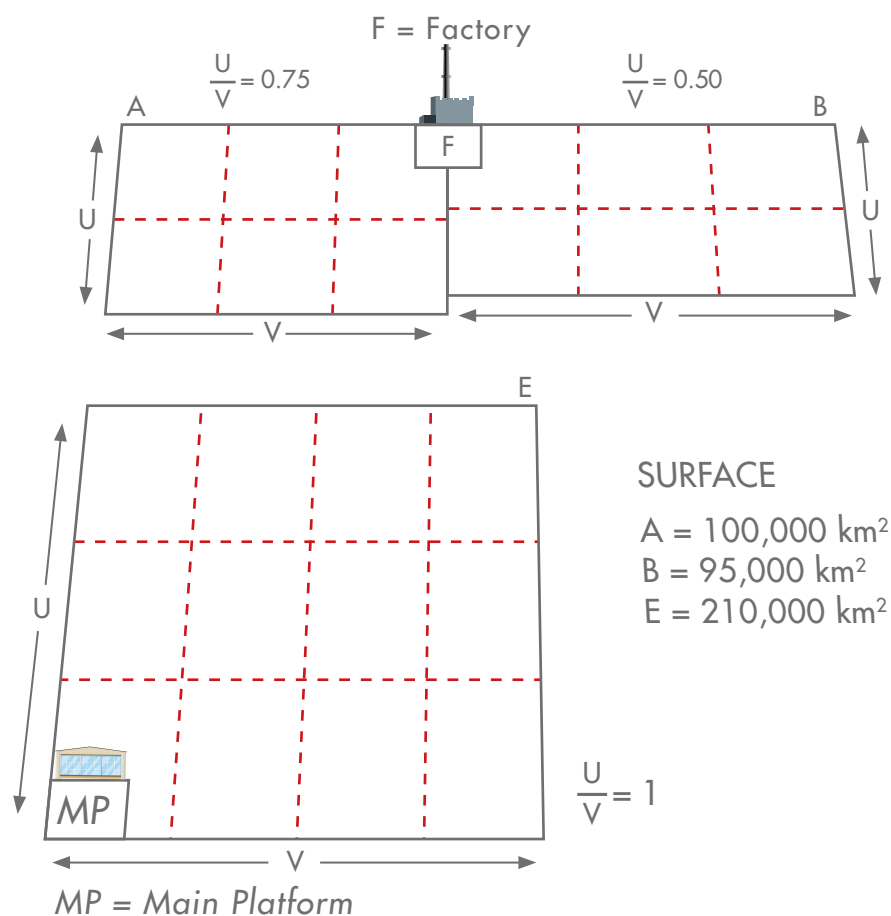
Cash&Carry stores are also called Supermarkets.



Each territory is structured into areas that require sales managers, who will be responsible for visiting mainly department stores. In A and B there are 6 different zones and in E there are 12.

Participants can decide whether this intermediate distribution network is to be composed of the company's own sales managers or the property of independent wholesalers. Of course, a mixed solution is also possible; therefore, for each of the market-territory in which you want to compete, the number of sales managers and/or wholesalers to establish must be decided.

The territorial structure is homogeneous, so each element (sales force and retailers) within it is equidistant from its adjacent elements. The map of the territorial structure is as follows:



5. DEMOGRAPHIC PROFILES OF PURCHASERS

Demographic profiles are very similar in all markets. It has been detected that buyers represent the 25% of total population in all markets.

It is also known that out of that 25%, the number of female buyers equals that of male buyers, meaning that 50% corresponds to each sex.

6. COMMERCIAL DISTRIBUTION

Your company must decide in how many of the sales areas (6 in A, 6 in B and 12 in E) to establish its sales managers, assuming that independent wholesalers are used at the remaining ones. This is the same for the three markets-territories.

Sales managers handle channel G and lead a sales team which acts on the T and S retailers. Managers have a fixed annual cost of 50,000 euros, not including the remuneration of the sales representatives (reps) responsible for channels S and T. The negotiation with the G channel is carried out by the sales managers.

You must decide the total salary cost for each sales rep for channels S and T and, together with the number of people needed for each of these channels (assuming company sales activities in these channels).

The same happens in country E, except that the set annual cost of a sales manager totals 60,000 dollars / year and no sales force recruitment is necessary, as only channel G exists in E and, as mentioned above, this is handled by the sales manager.

The sales organization, whether handled by independent wholesalers or in-house sales force, is identical for S and H products, and managers and sales reps are responsible for both products.



The number of salespeople recorded will always be per sales zone and it is assumed to be identical for each zone in which the company assigns a manager. Thus, the total amount of salespeople will be equal to the salespeople per area multiplied by the number of areas with sales managers.

7. WHOLESALERS

The independent wholesalers employed will charge 8.46% of the retail price for normal units they sell, and 6.5% of the retail price for those sold on promotion.

These percentages are the same for A, B and E and, as we will see later on, include the transportation cost from wholesaler to retailer. In the case of market E, the customs duty is also included.

8. SALES MANAGEMENT AND PHYSICAL DISTRIBUTION

Having the product physically available for buyers requires two separate kinds of actions: one called “commercial” and the other called “physical distribution”. Both should be properly coordinated and planned.

Commercial action is performed by salespeople that negotiate with retailers to make the sale and take orders, in addition to assuring the presence of products in stores. In large markets these salespeople have to be close enough to retailers, thus making it necessary to divide the territory into several areas (zones), each controlled by a sales manager.



Markets A and B each require the establishment of 6 areas and their respective sales force. Market E requires 12 areas and their respective sales force. Each company can decide whether the sales force will be independent wholesalers or in-house sales managers. Of course, the decision can be a mixed solution, in which case the number of sales managers and wholesalers have to be determined for each territory in which you wish to sell. Either way, all the areas belonging to the markets in which the company decides to operate will have to be covered.

Once the sale has been made, physical distribution comes in. The orders placed with each manager (or sales rep in the case of channels S and T), each one in its own area, have to be distributed from platforms (warehouses), having one or more of them per area. These platforms should be located in such a way that the sum of the costs of supplying them from the factory, keeping a certain inventory, and distributing to retailers is minimal, while assuring on time delivery and avoiding stockouts.

As a result of the above it follows that the total number of platforms may or may not coincide with the established sales areas. In our case the minimum number of platforms will coincide with the number of areas (one platform per zone) and the maximum will be 6 platforms per zone. The sales person responsible for each zone (whether a sales manager or a wholesaler) will proceed in such a way that all the orders taken are served from the platforms assigned to its area and will be responsible for placing the necessary orders with the factory so that their platforms are adequately supplied in time and quantity.

All platforms, whether they belong to a wholesaler zone or depend on a sale manager, are supplied from the factory and therefore transportation costs from the factory to all platforms will be borne by the manufacturing company. However, warehousing costs and transportation costs from platforms to retailers will be borne by the wholesaler or manufacturer, depending on whether they are located in areas assigned to wholesalers or sales managers.



The participating companies must proceed to plan both the commercial action (order taking) and the physical distribution necessary for the correct and timely delivery of orders. To do this, bear in mind that the 6 zones in which each territory is divided (12 zones in the case of E) are equivalent in sales potential and that therefore each has the same number of retailers of the different types of channel in each territory.

In order to manage average values, adequate and necessary for proper planning, in each area it is assumed that the corresponding retailers are uniformly distributed over the territory to be covered.

It is necessary to keep in mind that sales management and physical distribution are intertwined and that the delivery of products sold (either through wholesalers or sale managers) requires the existence of platforms, since without them sales would be void.

Companies can vary any of the logistics decisions in every business plan.

There are three alternatives for your company:

A. Having sales managers in each sales area of the market.

In this case, you have to develop the entire logistics plan.

B. Not having any sales managers in a market.

In this case you must plan vehicles (number and load capacity) up to platforms and number of platforms.

C. Having between 1 and 5 sales managers in markets A and B and between 1 and 11 in market E.

In this case, you must develop the entire logistics plan, although the costs of platforms and distribution to retailers will be borne by wholesalers in the areas in which they operate. If your company decides, for example, 3 platforms for each zone, you will also have to deliver the products to the 3 platforms that are in a wholesale zone.

Vehicles designated to operate from platforms to retailers only operate in areas controlled by sales managers.

While most companies use the trial-error-new trial method to make decisions concerning the number of vehicles and their load capacity, certain large companies apply geometric operational research methods to make optimal decisions.

Participants can also proceed by trial-error method since, on the one hand, the results inform whether they have had a theoretical stockout (the logistic stockout is theoretical since the logistic company supplies any additional vehicles needed, avoiding the actual stockout). This information indicates that we underestimated and have fallen short of vehicles and/or load capacity, and therefore we will know that we have to increase the means of transport used.

Reference unit transport costs from Factory to Platform (0.006 euros) and from Platform to Retail (0.007 euros) are given for guidance purposes. These must be compared to the actual costs incurred by the company which can be found in the results. If the latter are greater and the company did not incur in a logistic stockout, we can assume that the transportation resources may be excessive and should be reduced.

Those participants who are so inclined may apply a geometric and time analysis to arrive at optimal or near optimal decisions. Doing so requires using all the data included below, though there will always be some uncertainty since the solutions depend on the number of units sold and require accurate sales forecasts.

The study of physical distribution channels has taken on great significance, as it is seen as a matter of cost reduction that can improve price competitiveness and that is closely linked to the transaction channel.

9. PHYSICAL DISTRIBUTION - LOGISTICS

Physical distribution always involves three phases in territories A and B, and six in E:

	PHASES	
Territories A and B	1	Transportation from factory to platforms
	2	Storage of products at platforms
	3	Transportation from platforms to retailers
Territory E	1	Transportation from factory to the main platform in E
	2	Storage of products at main platform
	3	Customs
	4	Transportation from main platform to platforms
	5	Storage of products in platforms
	6	Transportation from platforms to retailers



Therefore, two kinds of documents for administrative management are issued at the platforms:

1. Orders from platform to factory.
2. Delivery notes to retailers, corresponding to the orders taken by the retailers through the sales force, either wholesalers or sales managers.

Costs of phase 1 in A and B are always borne by the manufacturing company, no matter whether the platforms are located in areas managed by wholesalers or by sales managers.

In market E phases 1, 2, 3 and 4 are borne by the manufacturing company. Customs duties will be paid by the manufacturing company if the products are sold in areas where it has a sales manager and, on the contrary, will be paid by the wholesalers when the products are sold in areas managed by them. The cost of phases 1, 2 and 3 is fixed, amounts to 0.024 euros per unit and includes freight charge, insurance, customs clearance and the costs of the main platform, but does not include the customs duty, 2% ad valorem, calculated after clearing customs and with the merchandise ready to be delivered to the platforms.

$$(\text{Standard cost} + 0.024 \text{ euros}) \times 0.02 = \text{Unit customs duty cost}$$

Customs duty cost should only be taken into account when the cargo destination is platforms dependent on sales managers. When the cargo goes to platforms managed by wholesalers, the customs duty is included in the 8.46% charged by the wholesaler.

If a company decides to operate in the E market, the simulator will assign a main platform at the port of entry.



10. FACTORY TO PLATFORMS TRANSPORTATION

Transport from the factory to platforms implies a cost that depends on the participating companies' decisions and will therefore be different depending on the case. In E this cost is the cost of transportation from the main platform (after clearing customs) to the other platforms.

First of all, the extension and configuration of each territory, as described on the territorial structure map, should be taken into account.

The following experimental data are available:

Average speed	Market A	Market B	Market E
Between factory and platform	90 km/hr.	90 km/hr.	
Between platform and platform	90 km/hr.	90 km/hr.	100 km/hr.
Between main platform and platform			110 km/hr.

Average loading and unloading time	Market A	Market B	Market E
Loading at origin	0.5 hr	0.5 hr	0.5 hr
Unloading at platform	0.25 hr	0.25 hr	0.25 hr

The uptime of each vehicle and its driver is 2,000 hours/year.

The mean time employed by every vehicle for each route is important and should be taken into account. It obviously includes loading at the factory and unloading at the platforms, plus the journey in kilometers from the factory to platforms, between platforms and back to the factory. The average time of a route is 8 hours, sometimes it is more and others it is less.

The vehicles to be used must all be the same in terms of load capacity, which is in kilograms, and in our case one kilogram equals to two units of product, both for S and H.

Participating companies do not buy the vehicles; rather, they are assumed to be rented from a logistics company that provides all necessary services, including drivers and logistics scheduling, but not the planning. Hence, participating companies should plan at the beginning of the year the number of vehicles they will require, as well as their load capacity, knowing that the logistics services company will charge the following for each vehicle, including all costs:

$$\text{Annual cost per vehicle in euros} = 24,000 + 1.2 \times \text{load capacity in kilograms}$$

The same formula is used for E but considering the euro-dollar exchange rate.

The vehicles load capacity can range from a maximum of 24,000 kg to a minimum of 1,000 kg. Your company must decide which capacity will be more appropriate.

In addition to the goal of minimizing transport costs, it is essential that all units sold (both S and H) be transported from the factory to all of the platforms (to both those run by wholesalers and those in in-house sales areas). If this is not achieved, it will be necessary to request additional vehicles from the logistics company, which, as they were unplanned, will entail a higher cost than those previously rented. This will, however, avoid stockouts and their negative consequences.

If the stockout occurs at some point in the chain, your company will be informed in the results in which markets it has taken place, and this will have meant that your logistics supplier has been forced to provide additional vehicles not anticipated in the planning.

Your company has to make the following decisions:

- Total number of platforms to logistically cover each territory, with the following possibilities:
 - A and B: 6, 12, 18, 24, 30, 36.
 - E: 12, 24, 36, 48, 60, 72.
- Number of sales managers:
 - A and B: between 0 and 6.
 - E: between 0 and 12.
- Number of vehicles to hire with the logistics company in each market-territory.
- Load capacity in kilograms per vehicle that is hired in each market-territory. Remember that each kilogram includes two units, both in the case of S and H.

As a guideline, a reasonable logistics transportation cost from factory to platforms is around 0.006 euros per unit delivered.

11. PLATFORMS MANAGED BY SALES MANAGERS

These costs will only be borne by the manufacturing company in the case of having sale managers.

Platform costs corresponding to areas managed by wholesalers are charged to them, and therefore are included in their commission per unit sold (8.46% for normal units and 6.5% for units sold during promotions).

The experimental data available are the following:

- Administrative costs (inventories, delivery notes, orders, etc.).
These are grouped and are directly proportional to the number of annual orders to the factory by a platform and are estimated at 0.65 euros per order for A and B, and 0.65 dollars for E (assuming a 1:1 exchange rate). Therefore, the exchange rate should be taken into account at all times.

Furthermore, these costs include the delivery notes issued by the platform to the retailers, this being 0.15 euros per order delivered to the retailers. In E this cost is 0.15 dollars (assuming a 1:1 exchange rate).

- Storage costs (space, insurance, cleaning, energy, etc.).
These costs will be directly proportional to the number of units that constitute the average stock throughout the year at the platform and have a cost of 2 euros per unit in A and B. In E the cost is 2 dollars, assuming a 1:1 exchange rate.
- Handling costs.
These costs account for moving cargo to, from and within the platform (including loading and unloading), and are directly proportional to the number of units annually handled by the platform (i.e. units sold in each area, divided by the number of platforms in that zone) at 0.0027 euros per unit in A, B and E (assuming a 1:1 exchange rate in E).

Platforms are assumed to be rented, which allows modifications every year.

Sales managers and reps salaries include sales and administrative costs and are independent from platforms costs.

12. PLATFORMS TO RETAILERS TRANSPORTATION

Transport costs from a platform managed by wholesalers to retailers are borne by the wholesaler and are included in 8.46% for normal units and 6.5% for units sold in promotion they earn.

In order to estimate the transportation cost from platforms located in areas with in-house sales force to retailers, the following experimental data are available:

Average speed	Market A	Market B	Market E
Between platform and retailer	60 km/hr.	60 km/hr.	80 km/hr.
Between retailer and retailer	40 km/hr.	40 km/hr.	60 km/hr.

Average loading and unloading time	Market A	Market B	Market E
Loading at origin	0.5 hr	0.5 hr	0.5 hr
Unloading at retailer	0.16 hr	0.16 hr	0.16 hr

The uptime of each vehicle and its driver is 2,000 hours / year.

The average route duration does not have to be 8 hours unlike in the case of transportation from factory to platforms. Given the proximity between platforms and retailers, the average route time can be any.

The participating company must decide on the number of vehicles and their load capacity, which does not necessarily have to be the same as those planned for transportation from factory to platform.

Services are contracted out to a logistics company that will cover, should a decision produce a stockout, the deficiencies in the company's planning by providing the necessary extra resources, though at a higher cost.

The annual cost per vehicle is the same as for those used between the factory and platforms:

$$\text{Annual cost per vehicle in euros} = 24,000 + 1.2 \times \text{load capacity in kilograms}$$

In E the exchange rate must be taken into account.

The load capacity of the vehicles can range from a maximum of 10,000 kilograms to a minimum of 500 kilograms.

Your company has to make the following decisions:

- Number of vehicles to be rented from the logistics company in each market-territory.
- Load capacity in kilograms of the vehicles rented in each market-territory. Remember that one kilogram is equal to two units of product.

It must be stressed that not every retailer has to be served, only those located in areas managed by in-house sales force and therefore served through platforms run by them.

In addition, keep in mind that retailers make weekly orders for the average number of units sold each week, assuming 52 weeks per year.

As a guideline, a reasonable logistics transportation cost from platforms to retailers is around 0.007 euros per unit delivered.



Values of zero may not be entered in logistics decisions.
If wholesalers are handling the transportation from platforms to retailers no cost will be charged for this concept.

13. SALES FORCE

This includes the sales representatives assigned to areas where there is a sales manager.

The company has to decide on the size of its sales force for channels S and T in markets A and B, together with its salary policy, that will consist of a fixed and a variable salary and can be different in the two markets. The range is:

Salary	Minimum	Maximum
Fixed portion	21,000 euros/year	42,000 euros/year
Variable portion	0.6%	1.2%

The fixed salary is a mandatory decision, whereas the variable is optional but, in case of deciding to have one, it must be a percentage within the above mentioned range.

The variable salary will be based upon normal units sold, will not include those sold through promotion.

Commissions will be calculated based on the price to the retailer, not on the retail price (MSRP).

Units sold through wholesalers are not subject to any payment of commissions to the company's sales representatives.

14. PRICES

Each company has to set the prices for each product in every channel and market where it decides to operate.

These prices are always the sales prices to the general public or final consumer (retail price, MSRP), and therefore each company's sales will equal these prices multiplied by the units sold in each channel and market.

Prices can obviously be adjusted every year and up to two decimal places may be used.



The simulator assumes that companies are operating in a channel with a product if a price has been set for said product. If no price is specified, the simulator will assume that the company is not commercializing.

15. ADVERTISING

There are several types of mass media in the markets, used in either a digital or traditional way, whose audience is limited to the territory itself.



Rival companies can use one or several advertising media in each market. The types of advertising outlets with their rates expressed in dollars and euros, by insertion/repetition, are the following in the two markets:

Media		A and B in €	E in \$
Display network and Remarketing <i>Banners/ads on websites</i>	DR	10,000	11,000
Press <i>Newspapers/Magazines: Print and online</i>	PR	15,000	16,000
Radio <i>Stations: Conventional and online</i>	RA	14,000	16,000
Social media <i>Closed communication platforms</i>	SM	15,000	15,000
Television <i>Channels: Conventional and online</i>	TV	18,000	20,000



You must specify the total annual number of insertions.

16. SCOPE (RATING)

The following table shows the percentage of buyers reached by one insertion/repetition of each advertising media. That is the scope.

Media		A and B	E
Display network and Remarketing <i>Banners/ads on websites</i>	DR	28	26
Press <i>Newspapers/Magazines: Print and online</i>	PR	48	31
Radio <i>Stations: Conventional and online</i>	RA	35	40
Social media <i>Closed communication platforms</i>	SM	30	29
Television <i>Channels: Conventional and online</i>	TV	60	61



Knowing how many people will be reached, over the 25% of population which are the buyers, you will have to decide how many times to reach each of them. That is the insertions/repetitions/frequency, the number of times that you want your advertisement to be seen.

All outlets overlap some other outlet. That means buyers, generally, are not only receptors of one single advertising media, but of two or more. The market study on coverage can allow you to know the percentage of buyers that has been reached.

17. BRAND POSITIONING

Your advertising agency presents you with different alternatives as described in the Appendix. Each of these ads/campaigns includes the description of a visual scenario and a text, the essence of the message to be conveyed, that can be used for audio-visual and printed media. Your company must choose one of them for each market and product that you wish to sell. The advertising campaign and budget can change each year.

The selected campaign or campaigns will be used, without distinction, on that media the company decides to place insertions on. The advertising campaign and budget can change every year.

There are three major consumer segments, which give rise to different buying attitudes and brand choices. The weights of each of them are 40%, 40% and 20%. It must be the companies that locate the segments and influence them according to their strategic plans.

If you commission the market research that gives you the positioning maps of competing brands, you will see two unlabeled axes that have to be interpreted. They are drawn up using the most advanced statistical techniques, specifically with the non-metric multidimensional scaling method, NMMS, in its Kyst computer version. The axes represent those attributes, or sets of attributes, which are valued by consumers. Their meanings require interpretation, as it would happen in real life.

A brand's image attributes usually cannot be changed in the span of one year, but due to the fact that the number of simulated years is reduced, the simulator allows the positioning to be modified every year, so as to maximize training value. If this were not the case, the trial and error process would not be possible with this particular feature of brand image.

Therefore, your company can change its positioning from one year to the next if you wish to do so, not needing to make additional marketing efforts.

In market E there are three relevant local brands that appear positioned on the map. They are called J, K and L.



These three brands are located in the three essential strategic spots of market E, which correspond to the three largest consumer segments.

In every other marketing aspect, the three brands, along with those having an insignificant market share, behave like a single company, as if these domestic brands were in close cooperation with each other.



If your company does not choose an advertising campaign on the decisions, the simulator will assign you a neutral, undefined positioning.

18. SALES PROMOTION

Competing companies may carry out specific promotional sales campaigns. To do so, they must decide what products, and in which channels and markets, they want to promote. In each of the cases the number of annual promotions must be decided, bearing in mind that every year a maximum of 3 promotions can be implemented in each channel for each product.

The next decision is the type of promotion that your company prefers. In a given year, you can only select one type of promotion per each channel. For example, if a company decides to do 2 promotions of the 3x2 type in market B, channel S supermarkets, both promotions will be of that type. Different decisions can be made for other channels.

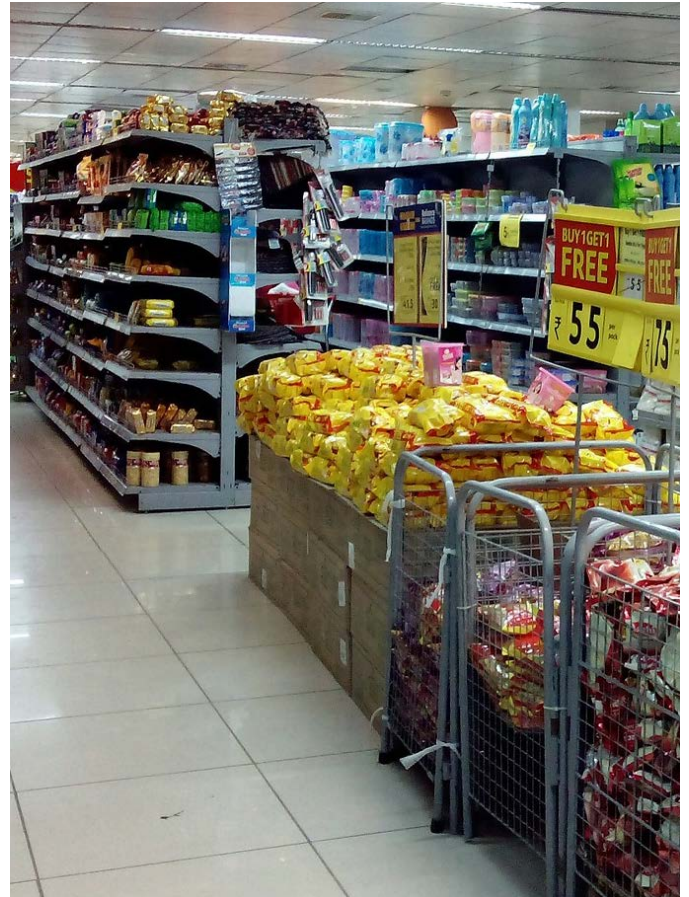
It is possible and allowed by the simulator for two or more participating companies to have promotions of the same type for a product in the same channel and market, but the simulator prevents these promotions from being carried out simultaneously, just as in real life. In assessing the sales that may result from any promotion, keep in mind that its effect lasts 3 weeks and that the simulation assumes 52 weeks. It should be clear that the promotion may not be accepted or that, on the other hand, it may be so successful that it exceeds all estimates. Experience and market knowledge will provide the necessary information for making these decisions.

There are eight different types of promotions:

1. Price reduction: This consists of a discount on the retail price that the company sets for a certain market, channel and product. The amount to be discounted must be stated on the decisions, in euros per unit. If the promotion is in E, the discount should be specified in euros, which the simulator converts to dollars according to the exchange rate in effect.
2. 3x2: The retailer will give the end consumer three units of S or H for the price of two at the time of purchase.
3. Drawing: In this type of promotion the participating company must specify the budget dedicated to each drawing. This amount, stated in euros, is for a single promotion and the simulator will multiply this budget by the number of promotions to be carried out in the channel.
4. Discount on next purchase: Upon purchasing one unit of product, the consumer receives a discount coupon that can be used on the next purchase. You must record on the decisions how many euros

are discounted on the retail price when the coupon is exchanged at the moment of the second purchase.

5. Direct promotional bonus: Instant gift of a purse when purchasing X units. This gift costs the company 1.20 euros. On the decisions you should state the number of units the consumer needs to buy in a single purchase in order to receive the gift.
6. Indirect promotional bonus: A gift obtained by accumulating points on different purchases. The decision to be made is the number of units the consumer needs to buy in order to claim the gift. The cost to the company is of 3.60 euros per gift. If you opt for this type of promotion, you must choose from three different alternatives (gifts):
 - A travel toiletry bag.
 - A towel.
 - A Pamela hat.



The simulator (which behaves just like the market) will only accept sufficiently attractive promotions. On the results the participating company can see which promotions have been accepted. Any rejected promotions generate no cost whatsoever, with the exception of drawings.



When entering decisions, you should indicate only those that are necessary for your promotion to take place. For example, in case of a 3x2 promotion it is not necessary to decide anything else, in case of a discount on next purchase promotion you have to decide the amount in euros that will be discounted, in case of a direct or indirect promotional bonus, the number of units that the consumer needs to buy to obtain the gift should be inserted. The only case in which the amount to be budgeted is necessary is for the drawings.

19. POINT OF PURCHASE ADVERTISING

The company must decide the budget to be allocated to posters, display stands, etc., to be placed in stores carrying their products. The budgets for products S and H must be specified separately.

The budget on POP advertising should be similar to a real world case, just like all the other decisions.

20. RETAILERS

The participating company must decide the margin granted to retailers. This must be done in each market, channel and product; that is to say, in all cases where prices have been set.



The margin for each unit sold should be stated on the decisions in euros, and not as a percentage. Although obvious, you should take into account that the difference between the selling price to the final consumer (MSRP) and the margin represents the selling price to your clients.



If your company does not specify a margin, it will be assumed that you do not want to operate in that channel, hence no sales will occur.

21. SHELF SPACE

The participating company must study how much shelf space it wants to occupy with its products, taking into consideration all factors that contribute to achieving a lower or higher figure.



If your company does not determine a shelf space in its decisions, it will be assumed that the company does not wish to operate in such channel and, therefore, will not sell.

These percentage-based decisions, made for each market, channel and product will be understood as the company's objectives subject to negotiation with managers of the different points of sale. Such negotiations are simulated by the simulator, so that the percentage of shelf space actually achieved will possibly be different from the one the company noted as its objective.



In the company's decisions, you must indicate a feasible objective as a percentage of the shelf space.

Therefore, there are two "percentages of shelf space".

One at the beginning and one at the end of the year. The one at the beginning is what your company sets as an objective and reflects in the decisions.

Then the simulator intervenes, emulating the corrections that the retailer makes throughout the year, according to your product's performance as compared to those of your competitors. These changes will result in an end of the year percentage of shelf space, which is what is reflected in a market research survey. It is recommended not to insert illogical percentage objective, with excessive growth, since it would be counterproductive, the retailer would react negatively. Mark coherent objectives and a feasible growth.

22. PRODUCTION FACILITIES



Production is undertaken by installing complete processing lines. These lines are made up of a set of components, assembled as a whole, of various types: mechanical, hydraulic, thermal and electric. They are fed by the necessary raw materials for complete production, including packaging of units ready for storage and sale.

While the lines basically operate automatically, two types of workers are required: Specialists and Operators. The former are mainly required to regulate and co-ordinate the components of the line and to control the process. Operators are basically needed to replenish raw materials and supplies, and to store and monitor finished products. The lines require a certain supply of electricity in order to operate.

Complete processing lines for S and H can be acquired and installed. Both lines are placed parallel to each other and it is assumed that the production facilities have enough surface area to accommodate up to 32 complete lines, of both S or H, or any combination of the two, as long as the total amounts to a maximum of 32.

S and H lines are not identical and therefore are not interchangeable, so if an S line is purchased and installed, only S products can be produced on it. The same is true for an H line.

Once a line is acquired and installed it cannot be eliminated and will remain installed until the end of the simulation, even if not used. Bear in mind the building work necessary for its installation. The purchase and installation costs (included the building work) are 1.5 million euros for an H and 2.1 million euros

for a S line. Throughout the simulation, S and H lines can be added every year, as long as the total does not exceed 32. Even though it is possible to add lines, they cannot be eliminated, although they can be left inoperative.

The production policy for each year must specify the number of new S and H lines to be installed and those that are activated and put into production, since it is not compulsory to use all the lines installed.

New lines are assumed to work at maximum capacity in the same year they are installed.



If your company does not activate the production lines on the decisions, it is assumed that you do not wish to put them into operation, hence no production will occur.

23. PRODUCTION CAPACITY

Every year the company must decide how many production shifts to use, with a maximum of 3. Each shift lasts 8 hours. Once the number of shifts is decided (this can be changed in each management plan), this is applied to all the activated lines. It is not possible to assign a different number of shifts to different lines.

The production capacity of each S line is 240,000 units per year per shift.

With regard to the H lines, the theoretical capacity is 600,000 units per year per shift.

The working year is 2,000 hours.

24. MAINTENANCE

A line's theoretical production can be reduced by two main reasons: breakdowns and readjustment and cleaning work. Occasionally, theoretical production speed can also decrease because of a lack of supply of raw materials and consumables.

For every 2,000 hours of line operation, an average of 70 breakdowns will occur. If the line is operated for two or three shifts (4,000 or 6,000 hours), the breakdowns increase proportionately. Repair of a breakdown takes an average of 2.5 hours.

Those 2.5 hours are approximately made up of the following:

1. 0.5 hours for detection and localization.
2. 0.5 hours for preparation.
3. 1.5 hours for the actual repair.



A breakdown does not only cause loss of time, it also means a loss of quality of the product being processed, leading to the rejection of these units. These lost units usually match the ones produced during the time required to detect and locate the breakdown. This applies to both S and H production lines.

Repairs are carried out by outside personnel, qualified technicians who intervene when necessary. Each breakdown requires the presence of 4 outside workers, with the only cost to the company being that of the hours employed by these specialists at the rate of 24.041 euros per man-hour. These costs are added to the standard production cost.

Cleaning, readjustment and routine maintenance tasks do not imply loss of time or product, except for those corresponding to a processing line module that we will call the main module (for both S and H lines). Its readjustment, maintenance and cleaning requires extraction from the line and reincorporation into the process after reconditioning.

This operation on the main module requires 4 specialists working 5 hours. These workers can be chosen from those who normally work on the line, since the whole process is halted during these 5 hours. This operation must be carried out every 40 line operating hours. This applies to S and H lines.

25. PREVENTIVE MAINTENANCE

Instituting a preventive maintenance system allows you to foresee approximately 70% of breakdowns throughout the plant. So, while breakdowns and their repair are not completely avoided, by being able to foresee them, you can anticipate their detection and localization with subsequent savings of time and product waste. It requires extremely technical work that can only be performed by highly specialized engineers, which handle the maintenance plans at an additional cost of 500,000 euros a year.

We should not forget that 30% of breakdowns are usually unpredictable. The decision to establish a preventive maintenance department can be made every year.



26. QUALITY CONTROL

Your company can acquire (from an R&D firm) and install a set of devices, instantaneous detectors for all types of breakdowns and/or alteration in the quality of a product being processed, even in the final stages. These detectors, called PY (Poka Yoke), stop the production process as soon as they perceive an anomaly and eliminate the need for quality control afterwards.

Since the exact affected point of the line is visually and audibly detected, the time wasted in detecting and locating the breakdown is saved and loss of quality of the product being processed is avoided. The net result is one by one quality control.



The PY decision implies an investment that, once made, is irreversible. Therefore, the decision can only be made once during the whole simulation, in the year chosen by the company. If your company decides to install the PY system, every line already installed will be affected, as well as any lines installed in the future. Hence, it is not feasible for only some lines in your factory to be equipped with the PY system.

This decision involves an investment that would increase fixed assets by 720,000 euros for each S or H line.

27. MAIN PRODUCTION MODULE

Operations for readjustment and cleaning of the main module can be performed outside the process, that is to say, without stopping production. This is done by having an exact duplicate of the module to be readjusted (called SMED) so one can be replaced by the other. The substitution can be carried out with such speed that the time it takes to disconnect the module being used and install the replacement is insignificant in terms of halting the process.

All this does not imply eliminating the readjustment and cleaning work on the module being used (that will take 4 specialists 5 hours each), but the task is performed outside the production process. Duplicate modules are somewhat special and the system requires re-engineering the lines acquired initially, as well as those to be added in the future. Because of this, the decision to purchase SMED modules is irreversible and conditions the purchases of future lines.



The complication of adapting the SMED to your lines forces the factory to be operational for at least one year before this system can be incorporated. Consequently, the decision to purchase replica modules can only be made from the second year of the simulation onwards.

Machinery manufacturers have managed to get the same SMED modules to work in both S and H lines and just one module can be used for both types.

Acquisition and adaptation of lines implies an irrecoverable investment in fixed assets of 3 million euros per duplicate module, even if the participants decide not to use this SMED system after acquiring it. Consequently, the decision to incorporate this system is made only once, and at any time during the simulation. The decision to use the system once acquired is made every year.

Maximum exploitation of the modules requires re-phasing the start-up of the lines, which implies an initial time loss. This loss is not taken into consideration in the units produced or standard cost calculations.

Once you have decided whether or not to incorporate the SMED system, you must then decide, and state in your management plan, how many of these modules are to be purchased, the timing of your acquisitions, how many you will put into operation and how many 4-specialist operator teams per shift you will dedicate to module substitution and readjustment work.

Throughout the simulation, if the company increases the total number of lines, the number of duplicates can be increased if considered convenient or necessary.

The operator teams, devoted to the SMED, are added to the processing line personnel. They are company employees working a full 8-hour day and cost the same as production specialists.

The price of a SMED amounts to 3 million euros. There is no difference in any of the above between S and H lines.

28. PURCHASING AND SUPPLIES

Every year the participating companies will specify the raw material units and operating supplies required to satisfy their production plans and, obviously, their sales forecast. These units, which we will call “raw materials”, represent a considerable variety of products necessary to feed and assure the correct functioning of the components which make up the production lines. They include, for example, lubricants, spare parts, etc., in addition to the actual raw materials, which also include containers and packaging.

The combined result of all these products necessary for the whole process to produce one unit is what we call a “raw material unit.” Each raw material unit represents all that is necessary to produce one unit of finished and packaged product. So, participants should decide on the number of these units to purchase so as to cover the units that the production process will consume. You also need to bear in mind the inventory you might have from the previous year.

Participants are informed of the raw material unit cost every year. The simulation Director can change this cost at his/her convenience. Nevertheless, the price given to each company can be modified through negotiation with suppliers. This negotiation, which is reproduced internally by the simulator, basically depends on the following:

- Number of units purchased.
- Negotiated payment terms.

Companies must make two decisions for H and two for S, assuming that they decide to manufacture and market both products. Regarding payment terms, there are 4 options:

- 0 months (Cash).
- 2 months (60 days).
- 3 months (90 days).
- 4 months (120 days).

The payment term decision can differ between H and S since the raw material units are different and not interchangeable, and will always be separate. Their unit cost will also be different, with each being provided separately each year by the simulation Director.



The final unit cost for both H and S is always assumed to include the transportation cost from the supplier to your facilities.

Suppliers invoice monthly. Note that the available raw material units must be able to cover not only the units manufactured, but also those lost as a consequence of quality control that usually originates from a component malfunction or breakdown in the processing lines.

A stock-out of raw materials has negative consequences. Not ordering supplies for your production requirements with enough time or in the correct form increases raw material costs by 15% for those units purchased outside the plan. Obviously, your factory will not shut down because a raw material is not available and the simulator will arrange it so that production can continue but at an additional cost of 15%. This increase results from the urgency of purchasing meetings, manufacturing by the supplier, transport, etc. Therefore, you must plan your requirements accurately for each period.



Even if your company does not buy raw materials, on the result sheets you will find the price you would have obtained with the payment conditions that you put on the management plan. This is a tentative figure.

29. OFFICE AND MANAGEMENT STAFF

The personnel of the company that you are managing is divided into 3 groups:

a.	Office and management staff (as fixed expenses)
b.	Direct production personnel (standard production cost)
c.	Sales representatives (considered a sales management)

The flowchart of the office and management staff is divided into four levels:

Office and management staff flow chart		
I	General management	1 person
II	Department Managers (strategic level)	8 people
III	Tactical decisions for each function	15 people
IV	Executors	32 people

The number of people and levels is the same for all participating companies.

The flow chart does not include personnel from the Preventive Maintenance department, if your company decides to establish one. These personnel and their cost, specified earlier, are considered as a fixed expense, since they do not intervene directly in production, and their task is exclusively the forecasting of breakdowns.

Level II corresponds to departmental managers. One of those departments is a small laboratory where product samples are analyzed (random sampling) for quality control. If you have decided to install the PY system, the laboratory work will mainly consist of checking and verifying the system detectors that in many cases are mini-laboratories. Each participating company will set its salary policy by assigning the remuneration of employees at each level (from the point of view of cost to the company). These have to stay within the following ranges:

Level	Minimum	Maximum
I	108,000 euros/yeavr	150,000 euros/year
II	72,124 euros/year	99,170 euros/year
III	51,100 euros/year	72,124 euros/year
IV	30,050 euros/year	42,075 euros/year

A single salary structure are established for all people within the same level.

General overhead expenses are 600,000 euros/year.

30. DIRECT PRODUCTION PERSONNEL

This refers to production line operators and includes supplying those lines, storage of raw material units, storage of end products, as well as maintenance and visual quality control.

There are two levels with the following remuneration ranges:

Level	Minimum	Maximum
Specialists	21,000 euros/year	29,450 euros/year
Operators	12,000 euros/year	16,830 euros/year

The number of people on the payroll depends on the number of activated processing lines and the systems installed and activated for the production process. Keep in mind that with regards to human resources, each shift has:

- For S Lines: 9 specialists and 32 operators per line.
- For H Lines: 6 specialists and 23 operators per line.

As stated earlier, all breakdowns are repaired by outside specialists who charge by the hour. Four people are required for any type of breakdown. The repair time will depend on the production system, and may fluctuate between 2.5 and 1.5 hours for each occurrence.

If you purchased and are using the SMED system, you will have to decide how many teams of 4 specialists are required to attend to it each shift, as mentioned before.

All of these expenses are under the labor component of standard production costs.

Overtime is not permitted and all personnel can only work one shift.

31. WORK ENVIRONMENT

You should keep in mind other expenses resulting from social and micro-environmental reasons your company could incur. Lack of motivation in the workplace or the inability to fulfill an assigned task produces harmful effects that are reflected in low productivity and work absenteeism. These effects differ for administrative, sales and direct production personnel.

Usually the lack of motivation in the first group is reflected as a high turnover of personnel that we define as the ratio between those people who leave the company every year and the total number of people on the payroll. This turnover depends not only on internal policies but also on the labor market and the rate of unemployment in territories A+B.

In the second group, loss of motivation has an effect on absenteeism that translates into low productivity and leads to higher production costs. In the first group, as a result of the need to replace the people leaving the company, high costs will appear that are a consequence of: search and selection of substitutes, adaptation time, payroll duplication for the time the two positions are maintained, etc.



The result sheets show the administrative staff and sales personnel turnover that has occurred during that period. People who quit the organization are immediately replaced.

32. SELECTION AND TRAINING

If the participating companies wish, they can choose the Training and Selection option that is tied to the internal promotion policy.

Every vacancy is filled by a company employee who had been working at the level immediately below the one vacated.

This policy aims to eliminate or reduce vacant positions, eliminating one of the causes that, to a certain extent, produces them: lack of promotion prospects. This applies to all personnel with the exception of unqualified operators, who will always be replaced by outside hires.



The use of internal promotion noticeably reduces costs inherent to personnel replacement. This method requires the creation of a department section made up of training experts and psychologists. Their main tasks, apart from other smaller ones, consist of applying a permanent training system by which each employee is trained in the functions and tasks at the immediately higher level, and in preparing employee aptitude and attitude profiles. This department, known as TS, whenever the decision is made to create it, involves an annual fixed expense of 722,000 euros. The decision can be made every year.

33. STANDARD COST

The standard cost, SC, of producing one unit of finished product is the sum of:

- Raw materials costs.
- Power costs.
- Personnel costs.

These three items differ for product S and for product H.

The raw materials costs are set by the simulation Director, corrected according to the terms and quantities negotiated with suppliers.

The simulation Director establishes the power costs.

Those corresponding to personnel depend on the previously mentioned production systems, reward policies and human resources.

These costs can vary each year and are reflected in the Operating Results as a weighted average, SCM, of the SC of the inventory held from the previous year and the SC of the current year.

34. INVENTORIES

An out-of-stock of finished products not only results in lower sales, but damages brand prestige and loyalty as well.

Inventory costs are assumed as being covered by the annual 600,000 euros of general overhead.

Inventory valuation is always calculated based on the weighted average standard cost, as explained earlier. In the case of raw materials this is done using the average cost, and for finished products using the total average of standard costs.

In order to calculate the inventory of raw material units, it is very important to bear in mind the units that are discarded during production as a result of quality control. Therefore, from the inventory of raw material units available (initial inventory + purchases) you must subtract the units used and not just those ones produced, since often the two do not match, the number of the units used generally being the larger of the two.

35. DEPRECIATION EXPENSE

Depreciation expense will vary depending on the number of shifts established every year. So we have:

Shifts	Fixed assets depreciation expense
1	8 %
2	10 %
3	12 %

The fixed asset accounts are as follows:

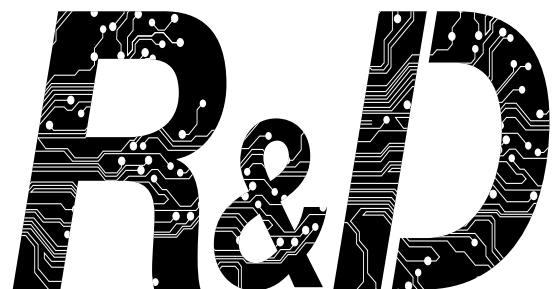
Production lines
R&D Poka Yoke
R&D SMED modules

Each year the percentages of depreciation expense is calculated based on the total figure and according to the number of shifts.

36. RESEARCH AND DEVELOPMENT

The industry you are in needs companies that are constantly innovating, so your company will devote 4.5% of its sales revenue to research and development.

Expenses derived from the activity that the company undergoes in this field will be reflected in the company's income statement under a specific item and will be different each year depending on the total sales revenue.



37. FINANCING

The simulation does not allow any increase in equity. The external sources for financing assets are: loans and credits from financial institutions and raw material suppliers. Investment in processing lines and their complementary elements are paid for in cash.

Delaying payments to raw material suppliers will generate financial resources whose benefits are offset by a higher raw material unit cost (resulting in the definition of "raw material unit" given in the "Purchasing and Supply" section).

38. CUSTOMER PAYMENTS

The payment terms given to customers are the same for all markets and channels, as well as for product S and for product H.

There are three options:

- Payment at 60 days (2 months).
- Payment at 90 days (3 months).
- Payment at 120 days (4 months).

The billing process is as follows:

The company issues monthly invoices to its clients for 1/12 of the annual sales (these products are not seasonal). These sales amount to the units sold under normal conditions and through promotions at the retail prices decided by the companies for both S and T. Retailer margins are paid immediately by the company. The same applies to promotional expenses, which are paid in the same month they are incurred.

39. FACTORING

Participating companies will have to decide if they wish to wait until the invoices from the customers reach their maturity date, or if they should be taken to the bank for factoring, so that greater liquidity is made available sooner.

In the first case, the entry in the “Clients” account on the Balance Sheet will always have a certain value, larger for longer payment terms. In the second case the entry in “Clients” will be zero.

If the factoring option is chosen, the bank will cash the bill at a specific discount rate, as set by the simulation Director. In the Income Statement, these expenses appear as “Financial Expenses from factoring.”



Another financial possibility is to discount a specified amount, as opposed to discounting the total amount of invoices. This is the same as having a discounting limit.



If you decide to use the factoring, you can set this limit by specifying it on the financial plan.

40. PERIODS - FISCAL YEAR

In order to handle the time variable, the fiscal year has been set equal to the calendar year. This is just a convention and is of no further importance except that the majority of companies' management plans also entail one-year periods that coincide with the fiscal year.

One could consider three month periods, or one month, or any length of time desired. It is merely a question of choice. The important thing is to experience the 4 steps of business performance: Analysis, Planning, Implementation and Control, which is what the simulator is for.

Having established a virtual period of one year, it is assumed that management plans are presented on January 2nd and the agreements with personnel, customers, suppliers and financial entities are also made on that date. Results are given on December 31st, being precisely this the date of the Balance Sheet and the Income Statement.

Therefore, the plan stated as a set of decisions goes into the execution phase on January 2nd. The results stated on the results reports correspond to December 31st.

Bank loans repayment plus recovery of fixed term deposits are made on January 2nd, that is to say, on the next working day after having received the Balance Sheet. The simulator will have frozen the amounts necessary to cover the outgoing funds for financial operations planned for January 2nd.



In case the Company has a cash shortage, the simulator will try to prevent an overdraft, since the highest interest rate charged by banks is that for overdrafts (remember this is an “in extremis” loan).

In order to do so, it will resort to financial investments. If the company holds any financial investments, these will be canceled to try to avoid the overdraft (this implies an early withdrawal fee).

Although obvious, it is necessary to point out that the company is not obliged to make financial investments. Interest rates are annual and internally the simulator uses monthly periods to calculate the interest rates applicable to financial operations. That is to say, each fiscal year is divided into 12 equal months.

Related to the above is the problem of lack of available Cash, which the simulator assumes is detected and suffered by the company the first month of activity in the fiscal year, just as the relationship between income and expenses becomes apparent.

Months are assumed to be equal and any real-life seasonality of products S and H is not taken into account by the simulator.

41. OPERATING CASH NEEDED

The operating cash needed “OCN” is the amount of money the company must have in cash at the beginning of a period in order to be able to implement the plan for the coming period.

The simulator considers your plan and estimates cash inflows and outflows. This valuation can be done beforehand by the company by following this rule:

1. 80 % of the new investments in fixed assets.
2. 50 % of the total of the following:
 - Total advertising and POP expenses.
 - Total sales managers expenses and sales reps fixed salaries.
 - General overhead.
 - Market research.
3. 15 % of cost of raw materials.

These percentages can be changed by the Simulation Director and if so, participants will be informed about it. The total of 1, 2 and 3 must be equal to or less than the available Cash. If such total amount is greater than the Cash, the company has two options:

1. Request a loan for the difference.
2. Reduce one or several of items 1,2 and /or 3 or eliminate any or several of them in order to make the total conform to the existing Cash.

The Cash amount to be considered is that of January 2nd, that is, the one received on December 31st, minus:

- Loan amortizations to be paid on January 2nd.
- Credit lines outstanding to be paid on January 2nd.
- Accounts payable.

and adding:

- Financial investments, if any, which are recouped on January 2nd.
- Fixed term deposits, if any, which are recouped on January 2nd.
- Accounts receivable.



If the OCN is higher than the theoretical cash on January 2nd, the simulator will request a loan for the company (since these are the additional funds needed) in an amount equal to the difference. If the OCN is lower, the company can use the Cash Surplus (CS) in financial investments and/or loan prepayment.

42. LOAN AND CASH & EQUIVALENTS

Every year companies will receive, together with market research results, their Financial Statements (Income Statement and Balance Sheet).

The Cash and Equivalents account can be negative, in which case it is obligatory to request a bank loan for that negative amount. This loan is granted in order to zero the account.

Apart from this mandatory loan, the company can decide to request loans at any time, even if the Cash and Equivalents balance is positive. Usually, such loans would be used for investing in fixed assets.

Companies should request loans (if needed) to comply with the OCN. If the company does not make a request, the simulator will do it.

Loan amortization will always be carried out so that each year, except the year in which the loan is requested (see Balance Sheet and Income Statement), the company returns a proportion of the total amount owed at the beginning of the year. The maturity of the loan and the applicable interest rate are set by the simulation Director.

If a new term or rate is set, it will be applied to the new loans and to the debt that the companies owe at that moment. The established amortization is done automatically by the simulator and is not to be specified on the decisions.

You can settle the debt, in part or in whole, before the set maturity date. Should you wish to do so, you must specify the debt to settle on the field labeled "Loan prepayment". To do this, the company must have Cash Surplus (CS) in the Cash and Equivalents account. The surplus amount can be used to settle any outstanding loans.



43. CREDIT LINES

Credit lines can be contracted with a bank in anticipation of Cash problems. The bank will charge the same upfront fee as for a loan, along with two interest rates, one to be applied to the amount not drawn and another to be applied to the amount of the credit drawn.

Credit lines last for one year but can be requested from the bank every year for the duration of the simulation.

The amounts drawn are stated on your Balance Sheet on December 31st and the simulator will automatically settle the amounts credited at the beginning of the following year. Therefore, if further funds are needed, another credit line or a loan should be requested.

44. FINANCIAL INVESTMENTS & FIXED TERM DEPOSITS

If you have a Cash Surplus (CS), this amount can be used to bankroll financial and/or fixed term investments, in both cases for a period of one year. The rate of return will depend on the amount invested/deposited and on the applicable interest rate, to be provided by the simulation Director.

The amount invested and/or deposited will appear in the Assets side of the Balance Sheet and can be cashed on the first day of the following year.



If Cash problems arise and liquidity is needed, the simulator will cancel the investment/deposit and the bank will charge the corresponding early withdrawal fee for this type of operation.

45. INVESTING IN OTHER COMPANIES (ACQUISITIONS)

A company with CS may, if it wishes, use those funds to purchase stocks and shares on the stock market. The securities you can buy will correspond to 3 real companies that actually trade on the market. The simulation Director will inform you of such securities, the applicable stock exchange and the market price of each share.

You can buy one or more of these securities by specifying the total cash amount you wish to invest, and not the number of shares. The simulator will perform this operation.

If, in the following year or years, you wish to buy again you can only do so by purchasing shares from another company whose shares you do not already hold. If you wish to increase the number of shares you already hold in a company, you must first sell them, and only then can you repurchase for the amount you wish. You can execute this double operation of buying and selling on the same “day”, that is to say in the same management plan.

You can sell shares whenever you wish, but if you decide to sell a security, you have to sell the total amount of shares you own of that particular security.

The profit or loss of the sale will be the difference between the purchase price and the sale price.

46. CURRENCY EXCHANGE

When introducing the decisions, amounts will always be expressed in euros. In E you should proceed in the same way and the amounts handled in dollars should also be expressed in euros, applying the corresponding exchange rate. The simulation Director will inform exporting companies of the exchange rate as needed.

As the participating companies (whether they export or not) are considered to be European, the Income Statement, the Balance Sheet and the Company Results in general will always be expressed in euros. It is therefore necessary to pay special attention when making your estimates and calculations since, for example, the advertising tariffs expressed in dollars do not vary but their value in euros does, according to the exchange rate. The same applies to the cost of branch facilities, internal transportation within E, etc.

47. INCOME STATEMENT AND BALANCE SHEET

In order to correctly analyze the Income Statement, it is necessary to understand that the invoicing or sales are expressed by considering the retail price (to the end consumer). Expenses, the retailer's margin and that of the wholesaler plus promotional expenses are deducted from this. These sales are the total units sold at retail price.

If your company requests a bank loan or credit you will see on the Balance Sheet that no repayment whatsoever is made that same year. The amount owed will be withdrawn from your Cash and Equivalents the following year, on January 2nd. Pay special attention to this fact when checking your Cash if you have outstanding loans or credits.

Therefore, when you are preparing your management plan for the following year, the simulator has already foreseen and frozen the funds payable on January 2nd.

The company's position regarding its competitors is indicated on the Result Sheets, after the Balance Sheet. This position is determined by the highest accumulated profit and its importance will be decided upon by the simulation Director. The simulation Director could give more importance to other issues, and you are informed accordingly if that is the case.

48. DEMAND

At the end of each fiscal year, results are given to the participants. The first two pages show the units of S and H sold. The last three entries indicate:

- A. Total units sold.
- B. Units demanded of your brand.
- C. Units of finished goods in inventory.



The units sold figure does not necessarily have to match the demand. This happens when the company's available production is lower than the demand achieved by it (stockout).

It could happen that, in an exceptional case, one of the participating companies does not even have sufficient production available to satisfy the demand generated by its promotions. In this case the market rejects those promotions that cannot be satisfied and only leaves those that can be supplied.

In addition to the total units sold and the demand obtained by your brand, the amount of finished goods in inventory at the end of the period is given.

4.9. MACROECONOMIC, FINANCIAL AND PRODUCTION FACTORS

Throughout this document the scenario in which the company will operate has been described. It has also been stated that the simulation Director will provide certain information. These are the figures you will receive and which should be kept in mind:

Per capita income A	€	27,000
Per capita income B	€	35,000
Per capita income E	\$	46,000
Exchange rate, euros per 1 dollar	€	1
Unemployment rate in A and B	%	7
Acquisitions		
Security 1	Market value	
Security 2	Market value	
Security 3	Market value	
Standard cost		
Raw material unit of S	€	1.65
Power cost per unit of S	€	0.25
Raw material unit of H	€	1.05
Power cost per unit of H	€	0.15
Factoring rate	%	5.5
Overdraft interest rate	%	20

Loans		
Maturity	Years	4
Interest rate	%	10
Upfront fee	%	1
Prepayment penalti	%	1
Credit lines		
Interest rate	%	7
Service fee	%	1
Upfront fee	%	1
Fixed term deposit		
Interest rate	%	2.5
Early withdrawal penalty	%	1
Financial investments		
Interest rate	%	2.5
Early withdrawal penalty	%	1



This economic and financial data are provided in the simulation interface and are shown in the General Information menu. It is very important to check this information whenever you access the simulation web as it may vary from year to year.

50. MARKET RESEARCH

Companies can carry out market research by commissioning this from a specialized institute. The type of information obtained, and the corresponding cost, expressed in euros, are as follows:



	Survey	Cost
1	Sales of S by market	3,000
2	Sales of H by market	3,000
3	Sales of S and H by company and market	36,000
4	Sales of S and H by company, market and channel	72,000
5	Promotional sales of S and H by company and market	89,800
6	Promotional sales of S and H by company, market and channel	107,000
7	Positioning of S brands by market	59,900
8	Positioning of H brands by market	59,900
9	Overseas advertising for S and H by companies from A and B, expressed in euros	36,000
10	Coverage of S and H advertising by company and market. Part of the target group reached, expressed in absolute figures	59,900
11	Number of advertising insertions for S and H by company	36,000
12	Advertising for S and H by E domestic companies, expressed in euros	36,000
13	Investment in PoP advertising for S and H by company, market and channel, expressed in euros	59,900
14	Number of sales managers and sales reps by company and market	24,000
15	Percentage of shelf space achieved for S and H by company and channel	72,000
16	Accumulated investment in fixed assets by company	49,000
17	Average overseas prices of S and H, expressed in Euros	24,000



	Survey	Cost
18	Total overseas sales of S and H by companies from A, B and E, expressed in units	36,000
19	Demand for normal units of S and H by company and market, not including promotional sales	110,000
20	Companies' S and H retail prices in euros by market and channel	Free
21	Companies' advertising campaigns for S and H by market	Free
22	Type of promotion, for S and H, by company and channel	Free
23	Companies' end of year net income	1,000
24	Industry's average production cost	10,000
25	Industry's average salary cost	5,000
26	Industry's end of year balance sheets	10,000
27	Number of logistics platforms by company and market	3,000
28	Companies' production shifts	1,000

Information given in surveys 1, 2, 3, 4, 5, 6, 18 and 19 is expressed in units.

Co-ordinates of positioning charts of surveys 7 and 8 are obtained using the non-metric multidimensional scale method NMMS.

Survey 21 indicates the advertising campaign by the number assigned in the appendix.

Information provided in survey 22 indicates the type of promotion by the number assigned in this document.

51. APPENDIX

ADVERTISING CAMPAIGNS

The campaigns that follow normally describe a television commercial. In the case of other media, the message and creativity would be adapted to the other type of communications media while maintaining the essential points.

SUNSCREEN

Campaign 1

In a cartoon, we see a small lizard sunbathing on a rock. As the camera zooms out, we see an aerosol can with sunscreen next to the lizard. The lizard's mother comes into view, crawling over the can, and leaning on the lid, rubs cream over the little one.

Voiceover: XXX sunscreen takes care of your little ones and the driest of skins.

Campaign 2

A boy beneath a beach umbrella looks dejectedly at another boy, out in the sun, happily playing nearby on the water's edge. The mother of the boy playing looks at him happily without worrying because she put XXX sunscreen on him.

Voiceover: "Don't fear the sun. Our new XXX sunscreen, with 56% less grease, takes care of you and those dearest to you".



Campaign 3

A castaway on a desert island is rescued and during an interview he's asked how he survived for so long. He answers: "With this," and he holds up XXX sunscreen.



Campaign 4

A woman on the beach. We hear her thoughts in a voiceover: "Oh, time to put sunscreen on the kids again." Image of the children running away and the mother running behind them with the sunscreen in hand, as if she were imagining it. Focus on the woman again, and the voiceover says: "Oh wait, it's XXX." She calls out: "Kids, sunscreen!" The kids start running, but they come back smiling when she says, "It's XXX."

Campaign 5

On a terrace at a beach bar, there are two men at separate tables eating a snack. One is red and scowling, since his whole body is burned. The other smiles at him while enjoying his snack.

Voiceover: "Don't let the sun ruin your vacation. Use XXX sunscreen and enjoy your days off with the best protection you can get."

Campaign 6

A dermatologist shows two pictures of skin, one damaged by the sun and the other in perfect condition, protected with XXX sunscreen.

Dermatologist: "Eight out of 10 dermatologists recommend XXX sunscreen. Its high iron and zinc content is the most effective way to protect your skin."

Campaign 7

A fire in a building and firefighters preparing to go in and evacuate it, but we see a man running ahead of them and go into the building while putting on XXX sunscreen. He comes out later without a single burn and holding a cat in his arms.

Voiceover: "The sun can give you second-degree burns, similar to what a fire can do. That's why we recommend using our XXX sunscreen."

Campaign 8

A group of people playing a game at a table. One of them reads a card and says: "Question: How can you enjoy the sun comfortably and safely?"

The other reply: "Under a beach umbrella," "under the water," "in an apartment"... Close-up of the game board, which shows a bottle of XXX.

Campaign 9

Man walking very stiffly, burned red by the sun, with a bathing suit on and a bag. Voiceover: "Don't end up like this. Use XXX sunscreen, now with better protection than ever."



Campaign 10

A person looking at a screen and typing. He starts speaking and we alternate shots of him and the screen, which shows carbon chains and their interactions.

"When you put a UVA particle on an organic compound like the skin, this is what happens."

We see the chain breaking apart. "If we put a thin layer of the three ingredients in XXX in between, we see something very different happen." The chain doesn't break.

The person stops looking at the screen, looks at the camera and says:

"Making this happen took a lot of effort. We've managed to protect your skin from the negative effects of the sun."

Campaign 11

A spaceship approaching the sun. Inside are four astronauts who are getting ready for a spacewalk. Three of them put on their spacesuits and helmets, while the fourth puts XXX sunscreen on.

Voiceover: "One out of 4 astronauts recommends XXX sunscreen. Its high iron and zinc content is the most effective way to protect your skin."

Campaign 12

Skin seen under a microscope. Little by little we see it flake, getting damaged and deteriorating.

Voiceover: "This is the damage caused by the sun that we can't see with the naked eye. Thanks to the new formula in our sunscreens, they protect your skin three times better than most other creams."



Campaign 13

We see a family having a picnic in a field. They're in the shade of a large tree from which they hang sunscreen, like fruit.

Voiceover: "Our XXX sunscreen, rich in aloe vera and vegetables oils, cares for your skin daily, restoring it and moisturizing it."

Campaign 14

Cartoon of a woman putting on cream and then rolling around in the sand afterwards.

Voiceover: "I love the beach. There's nothing I don't like about being at the beach and getting some sun, except having sand stick to my skin.

"XXX and say goodbye to sticky creams."

Campaign 15

A woman is getting her beach bag ready. Inside she puts a towel, sunglasses, fins... "Perfect, nothing can ruin my day at the beach now."

The same person appears later, returning home with a sunburn.

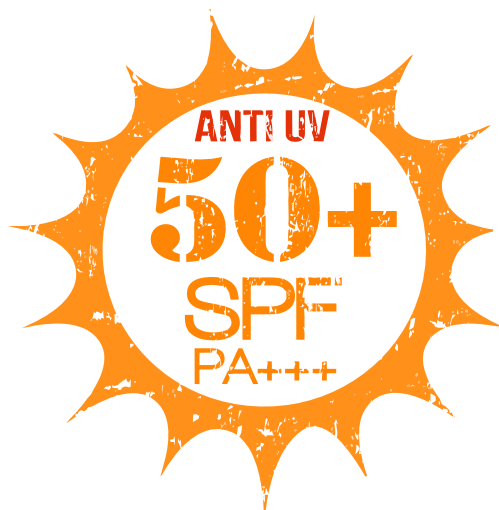
Voiceover: "Don't let anything spoil your vacation. Remember to bring XXX sunscreen along." We see her putting it in her bag.



Campaign 16

A series of images showing animals taking shelter from the sun, a lion lying in the shade, a bird under a branch, a dog under a table, etc. Finally, we see a person putting on sunscreen.

Voiceover: "At XXX laboratories, we've learned how to protect the skin naturally. Rich in aloe vera, our sunscreen restores skin while protecting it at the same time."



Campaign 17

Close-up of a tube of sunscreen showing the SPF (50+). Voiceover: "Do you know what this number means? This number multiplies how long it takes to get a sunburn. If you didn't use sunscreen, on average you would burn in 10 minutes. When you use our SPF 50+ cream, it will take 500 minutes to get a sunburn. That's why at XXX, we recommend asking your dermatologist how much protection you need to take care of your skin."

Campaign 18

A woman is speaking, and we see she's in an old, university-style lecture hall. Behind her is a projection on a screen.

"The sun is natural. It's natural for its harmful rays to reach us and damage the skin. And it's natural for us to cover ourselves so they don't harm us."

On the screen we see a person with a neoprene diving suit.

"But if we use the knowledge we have, we can protect ourselves with a few components that have been specifically developed."

Image on the screen of a bottle surrounded by chemical formulas.

Voiceover: "Technology for your skin."

Campaign 19

Images of a man flinching every time he gets a pat on the back or someone touches his shoulder or arm.

Voiceover: "You don't want this to happen to you when you come back from someplace sunny. To protect yourself from the sun, use XXX."



Campaign 20

We see parents trying to control three children who don't want to put on sunscreen. Focus in on the exasperated parents. Voiceover: "No one likes to struggle to put sunscreen on kids. Why go through it every time they come out of the water? Our XXX sunscreen is water resistant."

MOISTURIZER

Campaign 1

A person baking in the desert sun. Focus in on his happy, surprised face when he finds some moisturizing cream. He puts it on, and we see an oasis around him.

Voiceover: "XXX moisturizer. For maximum freshness wherever you are."



Campaign 2

A woman swimming and relaxing in a pool, turning over to do the backstroke. New image and we see that it was all in her imagination, since she's in fact sitting on a chair. Her friends, baking in the heat, give her a strange look and ask: "How can she stand this heat?" Focus in on the moisturizing cream next to her.

Voiceover: "XXX cream, as refreshing as a nice dip in the pool."



Campaign 3

A mother smiles and takes her son by the hand. They both run into the pool, laughing. As they go under the water, we see a tube of XXX-brand cream, sinking as air bubbles rise around it.

Voiceover: "Because we like taking care of you and your family, with our new and refreshing XXX, you don't have to worry about a thing, since it repairs and moisturizes your skin naturally."

Campaign 4

A young woman is walking on the beach, bending over every now and then to pick up seashells. Suddenly, while straightening out, she sees an object glistening ahead. Smiling, she walks up to it. It's a tube of XXX moisturizing cream in between some seashells. When she picks it up, a wave washes over her shins.

Voiceover: "New XXX moisturizer, more refreshing than ever."



Campaign 5

A desert full of sand. A tube of XXX moisturizing cream suddenly falls down. As soon as it hits the ground, it digs itself into the sand a little bit, and vegetation, vines and huge leaves sprout from the ground and start growing around it. The camera zooms out and we see an oasis in the middle of the desert, right where the tube of XXX fell.

Voiceover: "XXX moisturizing cream has all the vitamins and minerals you need to keep your skin moisturized all day long. Its natural active ingredients provide essential nutrients that care for your skin."

Campaign 6

A woman in a white lab robe shows two skin types on a virtual display. One is moisturized and the other shows the kinds of attacks that can damage skin.

Voiceover: "Every day the skin is attacked in ways we can't see. That's why we've developed a formula rich in vitamins E and C that protects and moisturizes your skin. It also has various minerals that help restore damaged skin. Take care of yourself, with XXX moisturizer."

Campaign 7

We see a tube of XXX moisturizing cream atop a rock covered with moss. Birds and other forest sounds are heard in the background.

Voiceover: "We like what's natural, which is why we make every effort to make the best product for your skin, with thyme oil extract and aloe vera, we've created a new formula for XXX moisturizing cream."

Campaign 8

Using a special animation effect, we see how when the lotion is spread on the skin, it moisturizes even the deepest layers, healing and helping regenerate damaged cells.

Voiceover: "The collagen and vegetable oil formula in our XXX moisturizing cream gives skin the vitamins and nutrients it needs to regenerate damaged cells, making skin feel fresh and smooth."

Campaign 9

A blazing sun. A young man with no shirt on is slumping in a chair on a small balcony, leaning against a table. Lying next to him is a dog, bearing the heat and panting.

A woman leans out the glass door and smiles. She goes inside the home and comes back with a bowl of water for the dog, who lifts its head and licks the air, thankful for the water. She then turns and stand behind the man, and smiling, puts some XXX moisturizing cream on his back.

The dog and man look at each other, smiling and thankful.

Voiceover: "XXX moisturizing cream takes care of you and your loved ones, protecting your skin every day."

Campaign 10

We see images of a perfect summer and a group of friends, surfing, sunbathing, eating on a terrace, etc. Between the images we see shots of the people putting on the moisturizing cream. Voiceover: "To enjoy summer, always moisturize your skin with XXX."

Campaign 11

A woman is taking a nice shower. She's humming along to a happy tune as she cools off under the water. Suddenly she looks up and reluctantly exits the shower. Wrapped in a towel in her room, she puts on XXX moisturizing cream and smiles again.

Voiceover: "Since you can't spend the entire day in the shower, XXX moisturizing cream helps keep your skin fresh 24 hours a day, moisturizing and protecting it."



Campaign 12

Voiceover: "If thinking about refreshing things brings this to mind..." We see images of a soft drink in a glass with ice cubes, watermelon, a pool, etc. on a background with different lively colors. "It's because you know..." We see a picture of the moisturizing cream.



Campaign 13

A man holding a baby in his arms, the baby won't stop crying and the father tries to hold him up against his neck, but the baby squirms and tries to move away. The mother then appears, taking the baby in her arms and holding it against her neck. The baby places a hand against the mother's cheek and instantly relaxes. The confused father looks at the mother, who smiles and points to a bottle of XXX cream on the shelf.

The next day the father is holding the baby, who is nestled against his chest.

Voiceover: "They'll notice it, and so will you. XXX moisturizing cream soothes and moisturizes skin, making it fresh and soft."

Campaign 14

A man with a lab coat rolls up his sleeves. The camera zooms in and, using special effects, seems to enter the scientist's skin, where we see all the layers of the skin and how water, nutrients and minerals are constantly refreshing it and rejuvenating it.

Voiceover: "With our formula rich in vitamins, minerals, water and nutrients, XXX moisturizing cream nourishes, regenerates and moisturizes your skin from the inside out. Its innovative four-level active shield will keep your skin safe from damage."

Campaign 15

Voiceover: "Don't forget to moisturize your body." We see hands taking a lunch pail with watermelon, a bottle of water and a jar of moisturizing cream. "To do that, remember to take XXX cream with you."

Campaign 16

A woman with her eyes closed puts her arms in the pool while sitting on the edge and moves them slowly. She then sticks in her legs and wets her shoulders, and lastly her face. Focus on her wet face, but when she opens her eyes, we see she's inside, sitting on a chair, holding a jar of the moisturizing cream.

Campaign 17

A mother putting moisturizing cream on her son. Voiceover: "Protect your skin and keep it moisturized with XXX cream."



Campaign 18

Inside a spaceship, with an astronaut doing a space walk outside. Inside is the captain, who is talking with ground control on the radio.

Captain: "All parameters normal."

Ground control: "And the humidity?"

The captain looks at a humidity gauge.

Captain: "Ten percent... but it's not a problem."

An astronaut throws him a jar of cream, which floats toward him. He grabs it with his hand. Close-up of a fine mist coming out of the jar and falling on the captain's face.

Voiceover, radio-style: "XXX, developed to give you the best results, even in tough situations."

Campaign 19

We see a woman with a cut aloe vera leaf, rubbing its liquid on her leg. Voiceover: "Imagine that feeling, only much better." The aloe vera leaf turns into the jar of moisturizing cream. "XXX cream, the most natural of creams, with aloe vera."

**Campaign 20**

A woman talking in her room while putting things in a backpack. "I take only what I need. That's why I never forget my moisturizing cream. It protects and repairs my skin like no other, making me feel fresh wherever I go."

© Praxis MMT. Experts in simulations. All rights reserved.

Copyright Praxis MMT Corp. All rights reserved.

www.praxismmt.com

MMT Simulations is a registered trademark of Praxis MMT Corp. This document is subject to change without notice.

License

The contents of this document are subject to the MMT Simulations License ("License"). This document can only be used in compliance with the License. You may obtain a copy of the License at www.praxismmt.com > License request > License Conditions.

Legal Notice

Praxis MMT Corporation is not responsible for: -the entry of data into the simulator -errors or omissions in the entry of data -the consequences these errors or omissions may have on your results -breakdowns of servers or telephone companies and the consequences they may cause.

Praxis MMT can't guarantee access to the web, since this is subject to deficiencies and defects caused by incorrect use, bad internet connection, accidents or external causes, such as: server crashes, hardware and/or server deficiencies, difficulty in accessing the server, damaged cables or antennas, interruptions in the connection, power outages, etc...

Praxis MMT Corporation is not responsible for keeping the user's passwords and access codes secret, or for the consequences of a third party knowing and using them, with or without the user's permission.

Further reproduction prohibited without permission.